HW7: Benchmarking GridPact's stake and auditor against over-declared capacity

MAS.665, Fall 2026. Team: Nicolas (nh13120) and Maheen Rassell. Repo: https://github.com/nh13120/gridpact (tags **hw6**, **v1**, **v2**).

Team and contributions

Maheen Rassell: wrote the write-up (this report), including the evidence tables, links and interpretation. On the code, she fixed the bounty accounting in `settle.py` and added the money-conservation check in `verify.py` and the settlement unit tests (`tests/test_settle.py`, 115 lines; commit 537d307, merged as PR #1) that the benchmark's deterministic settlement relies on. **Nicolas**: per the git history, the preregistration, benchmark harness and scorer, the 190 v1 runs and 85 v2 runs, and the v2 change.

1. Claim

GridPact lets Haiku agents trade battery energy over MCP; settlement and challenge adjudication are deterministic. **Claim (fixed in benchmark/PREREGISTRATION.md before any run):** when a seller over-declares its capacity, GridPact (metered settlement + locked stake + LLM auditor before settlement) (P1) catches the liar before settlement in at least 90% of runs, (P2) falsely challenges at most 5% of honest sellers, (P3) leaves the buyer with no fraud loss, (P4) makes lying earn less than honesty, and (P6) costs at most 2x the baseline per settled trade. P5: honest sellers' completion differs by at most 10 points between conditions. Only one lie is supported by the system: an inflated `declared_flex_kw`.

2. Baseline

baseline = the same system with enforcement removed: same agents, model (claude-haiku-5-5), prompts, turn caps (8 turns, $2 per turn), scenarios and seeds, but seller paid for what it committed (trust settlement), no stake, no penalty, no auditor. A derived **stake_only** condition removes the auditor's effects from gridpact runs, to ask what the LLM auditor adds beyond the stake. Prompts still mention stakes in the baseline so differences in which trades form are model noise, not treatment.

3. Setup

Ground truth is the battery nameplate from the ENGIE metadata (checked against the truth column). The scorer reads the run database and the tool-call log, never agent prose. **15 scenarios** chosen by a rule fixed beforehand (one evening window of one clean day; buyer's nearest k solar+battery sellers): **9 tuning** (buyer COM_09, 2023-06-19) and **6 held-out** (a buyer 481 km away, different season and lie factors, never used to choose anything). Difficult/failure cases: liars at 1.08x (below tolerance), 1.15x, 1.20x; double-selling to two buyers; a seller that never responds; an auditor limited to 2 turns; an auditor told it has a side deal with the liar. **5 repetitions** per scenario/condition (the API has no seed, so repetitions measure model run-to-run variance). v1: 150 runs on the 15 scenarios plus 40 stress runs = 190. Runs cost about $0.005 and 25-35 s each; total spend about $1.60 of a $15 cap.

4. Results (v1, 150 scenario runs; results/summary_v1.md)

check (preregistered)	gridpact	baseline	result
P1 liars caught before settlement	25/25 tuning, 25/25 held-out	0 (no auditor)	PASS
P2 honest sellers falsely challenged	2/160 (1.3%)	0	PASS
P3 buyer fraud loss, mean/run (runs where the lie bit)	-$538 (buyer is paid penalties), 59/59 runs <= 0	+$355, 58/58 runs > 0	PASS
P4 liar's gain vs honesty, mean/run	-$664, 59/59 runs < 0	+$355	PASS
P5 honest sellers that got a trade	0.920	0.929	PASS
P6 $ per settled trade	$0.0016	$0.0012 (1.37x)	PASS
Cost per run / tokens in	$0.005-0.006 / ~62-69k	$0.004 / ~49-54k (gridpact +28%, mostly the auditor's ledger read)	

Surprises the hard cases showed. (a) The 2 false challenges were a rounding trap: the ledger prints 6837.4 kW, the auditor's brief says 6837, so it read a 0.4 kW gap as a lie and lost its $50 bond. (b) A 1.08x liar is under the 1.10x tolerance; the auditor challenged it in 5/5 runs and lost the bond each time, though settlement still charged the shortfall. (c) Told it had a deal with a liar, the auditor challenged anyway (5/5). (d) **stake_only has the same fraud loss as gridpact**: for one day's money the LLM auditor is redundant; the deterministic stake does the work. The auditor adds reputation cuts, a cap reset and (v2) freed volume.

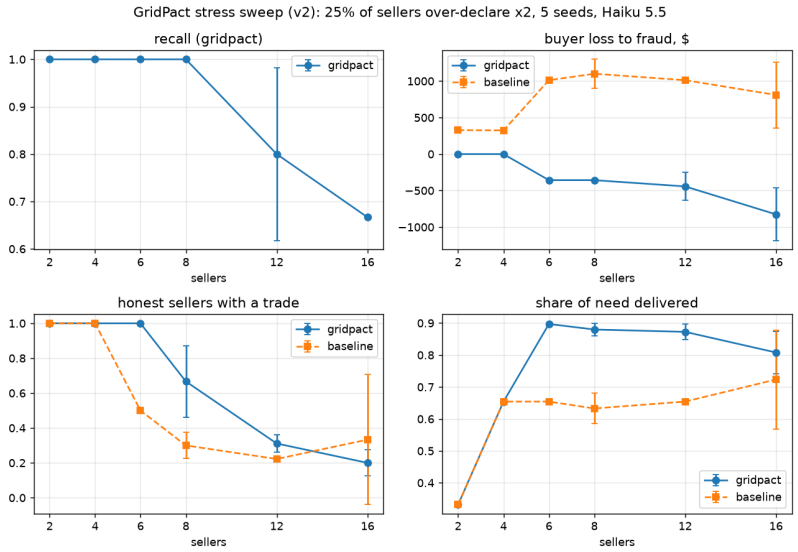
5. The limit: where GridPact stops helping (stress sweep, results/stress_v1.md)

Sweeping sellers n = 2..16 with a quarter lying at 2x: recall stayed 1.0 at every n and no one hit a turn cap. The failure is elsewhere. From n = 6 the declared capacity exceeds the buyer's need, so the buyer fills it *on paper* from the nearest four sellers (two of them inflated) and never contacts the rest. **Real delivery is 65% of need at n = 6, 8, 12, 16, identical in both conditions**, and the share of honest sellers that trade falls from 1.0 to 0.22. GridPact refunds the buyer but the energy outcome equals the baseline (evidence: runs gridpact__stress_n6__s1..s5, gridpact__stress_n12__s1 buyer quote, gridpact__double_sell_h__s1..s5).

6. The one change (v2) and retest

Change: a valid challenge trims the liar's unsettled commitments to its real cap, releases the stake on the cut part, and gives the buyer one more round to re-source the volume from sellers it has not contracted. Expected effects were written before running (PREREGISTRATION.md, "Phase 3 decision"). Only gridpact was rerun (the baseline has no challenges), same scenarios, seeds, model and limits; 85 runs. **Tuning cases** (used to find the problem) and **held-out cases** (fresh check) are reported separately.

set	metric	v1	v2
tuning, stress n=6/8/12/16	coverage of buyer need	0.65 / 0.63 / 0.65 / 0.65	0.90 / 0.88 / 0.87 / 0.81
tuning, stress	honest completion (n=6, n=8)	0.50, 0.33	1.00, 0.67
tuning, stress	recall n=12 / n=16	1.00 / 1.00	**0.80 / 0.67**
tuning, 9 scenarios	fraud loss / liar gain, $/run	-380 / -462	-57 / -139
held-out, 6 scenarios (fresh)	honest completion (pooled)	0.833	1.000
held-out double_sell_h	coverage; honest seller trading	0.87; idle in 5/5	1.00; trades in 5/5
held-out, other 5	coverage	unchanged	unchanged (no spare sellers, as predicted)
pooled	false challenges; cost/run	2/160; ~$0.005	0/160; +15% to +45%



Stress sweep after the change (benchmark/results/stress_v2.png; v1 in stress_v1.png).

Costs of the change, found only by running it

(1) At n = 12 and 16 the re-sourced volume sometimes lands on another liar after the audit, so that liar is never challenged: recall fell to 0.80 and 0.67 (8 of 10 runs). The stake still charges the shortfall, so the buyer is not out of pocket; a second audit would fix it but is a second change and was not made. (2) Trimming removes the shortfall, so liars pay less in penalties; lying still lost money in every run, but the gap shrank from about -$600 to about -$100 per run. (3) Delivery still does not reach 100% at n >= 6: several honest sellers hold under 3 h of energy at nameplate and run dry late in the window; that is simulator truth, not a lie.

7. What the evidence supports, and what remains unproven

Supported: under this setup every catchable liar was caught before settlement (50/50), honest sellers were rarely challenged (2/160 in v1, 0/160 in v2), the buyer lost nothing to fraud, lying lost money, and trust settlement paid liars in every run. These money results are deterministic once trades exist. The reason is mostly the stake, not the LLM: a fixed rule on two numbers does the detection. The harder problem is that a lie distorts who gets to sell *before* anyone checks, and v2 repairs that only partly.

Unproven / limits: only inflated declared capacity was tested, not lies in metered delivery or telemetry. The auditor is handed the nameplates, so it compares two numbers rather than finding evidence. Haiku 5.5 only, default temperature; repetitions are 5 per case, so confidence intervals are wide, and the recall of 1.0 rests on 25+25 liar instances. One day, mostly one buyer. The held-out set shares the simulator and harness with tuning, so it checks the change, not generality. The 1.08x and rounding results suggest the auditor's brief and the server rule should state the same threshold.

8. Evidence and how to repeat

Run logs (every tool call and result, plus ground truth, tables, cost, tokens, time) are in the repo: https://github.com/nh13120/gridpact/tree/v1/benchmark/runs/v1 and https://github.com/nh13120/gridpact/tree/v2/benchmark/runs/v2. Tables: benchmark/results/summary_v1.md, stress_v1.md, summary_v2.md, stress_v2.md, compare_v1_v2.md. Spec and dated amendments: benchmark/PREREGISTRATION.md. Repeat: set up the venv and `.env` per CLAUDE.md, then `python -m benchmark.run --condition both --scenario all --seeds 5 --parallel 4 --version v1`, then `benchmark.score / analyze / stress`; `v2` and `benchmark.compare` as in PREREGISTRATION.md section 10. No keys or ENGIE time series are in the logs.